HETEROGENEOUS ARCHITECTURES And PROGRAMMING MODELS

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Heterogeneous computing refers to systems which use more than one kind of processor or cores to maximise performance or energy efficiency. The processors incorporate specialised processing capabilities to handle particular tasks.

Heterogeneous architectures break away from the traditional evolutionary processor design path, posing new challenges and opportunities in high-performance computing. Heterogeneous computing will continue to add many cores and hardware features such as transactional memory, random number generators, scatter/gather and wider single-instruction multiple-data/advanced vector extensions, ensuring synergies with big data, mobile and graphics markets. It has the potential to achieve greater energy efficiency by combining traditional processors with unconventional cores such as custom logic, field-programmable gate arrays (FPGAs) or general-purpose graphics processing units (GPUs). Instead of using just a single CPU or GPU, heterogeneous architectures add an application-specific integrated circuit (ASIC) or FPGA to perform highly dedicated processing tasks. Heterogeneous architectures achieve performance gain by parallelism instead of clock frequency.

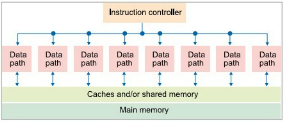

Fig. 1: SISD architecture

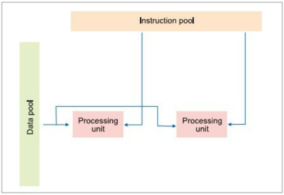

Fig. 2: MISD architecture

Parallelism

There are four levels of parallelism in hardware: single-instruction single-data (SISD), single-instruction multiple-data (SIMD), multiple-instruction single-data (MISD) and multiple-instruction multiple-data (MIMD).

SISD. In SISD, there are tens to hundreds of data paths, which all run by one instruction stream. General-purpose host CPU executes the main application, with data transfer and calls to the SIMD processor for compute-intensive kernels. SIMD has dominated high-performance computing (HPC) since the time of Cray-1 supercomputer.

SIMD. SIMD performance depends on hiding random-access memory latency, which may be hundreds of cycles, by accessing data in big chunks at a very high memory bandwidth. Data-parallel feed-forward applications are common in HPC. Large register files are provided in data paths to hold large regular data structures such as vectors.

MISD. It is a type of parallel computing architecture where many functional units perform different operations on the same data. Pipeline architectures belong to this type, though a purist might say that the data is different after processing by each stage in the pipeline. There is a sequence of data transmitted to the set of processors. Each processor executes a different instruction sequence.

MIMD. MIMD class of parallel architecture is the most familiar and possibly most basic form of parallel processor. MIMD architecture consists of a collection of 'n' independent, tightly-coupled processors, each with a memory that may be common to all processors, and/or local and not directly accessible by the other processors.

Two subdivisions of MIMD are single-